

Chapter 6: Anatomy of Flowering Plants

Monocot Root

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Class Notes · Topic 3/8

Same 4 regions as dicot root

Vascular bundle radial, exarch, polyarch (>6 patches)

Pith large & well developed

No cambium → no secondary growth ever

Epiblema, Cortex & Endodermis

As in the dicot root: a single-layered epiblema bearing unicellular root hairs, a wide parenchymatous cortex with intercellular spaces, and a single barrel-shaped endodermis with Casparian strips on the radial and tangential walls.

Pericycle

A single layer of thin-walled parenchymatous cells next to the endodermis, as in the dicot root.

Vascular Bundle

Radial, with **exarch** protoxylem, but here there are usually **more than six xylem patches** — a condition called **polyarch**.

Pith

Unlike the dicot root, the pith in a monocot root is **large and well developed**, occupying the centre of the stele.

No Secondary Growth

Monocotyledonous roots do **not** undergo any secondary growth, because of the lack of a vascular cambium in between the phloem and xylem — the pericycle here never differentiates into a cambium ring.

Dicot Root vs Monocot Root – Quick Contrast

Feature	Dicot Root	Monocot Root
Xylem patches	2–4 (mono/oligoarch)	>6 (polyarch)
Pith	Small, inconspicuous	Large, well developed
Secondary growth	Occurs (cambium ring forms)	Never occurs

Monocot Root at a Glance

- Radial vascular bundle
- Exarch, polyarch (>6 xylem patches)
- Large pith
- No secondary growth